



EFAST-2026

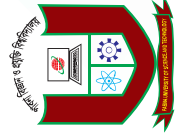
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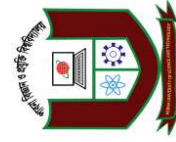
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District-Level Analysis of PM2.5 Trends and Associated Life Expectancy Loss in BangladeshMasura Mollika Niti^{1*}, and Md Kamrujjaman²¹Department of Applied Mathematics and Data Science, Asian University for Women, Chattogram, Bangladesh²Department of Mathematics, University of Dhaka, Dhaka-1000, Bangladesh

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Abstract

Air pollution has emerged as a critical environmental and public health concern in Bangladesh, yet comprehensive district-level assessments remain limited. This study examines long-term spatiotemporal patterns of fine particulate matter (PM2.5) across all 64 districts of Bangladesh from 1998 to 2020 and the associated health burden, measured as loss of life expectancy (LLP). District-level PM2.5 and LLP estimates were obtained from the Air Quality Life Index (AQLI) dataset, in which LLP is derived from the empirically estimated PM2.5 to mortality concentration response function and represents the life expectancy that would be gained if long-term PM2.5 were reduced to the WHO guideline of 5 $\mu\text{g}/\text{m}^3$. Temporal trends were assessed using descriptive time-series analysis at national and division levels; spatial variation was evaluated through district- and division-level comparisons, distributional summaries, and ranking; and district-level change analysis quantified increases over the study period. Averaged across the 64 districts, national PM2.5 rose from approximately 45 $\mu\text{g}/\text{m}^3$ in 1998 to over 73 $\mu\text{g}/\text{m}^3$ in 2020, while mean LLP increased from about 3.9 to over 6.7 years. Because LLP is a monotonic transformation of PM2.5 under the AQLI framework, districts with the highest concentrations also bear the greatest life expectancy loss. Substantial regional variation was observed, with the Dhaka division exhibiting the highest pollution and LLP and the Rangpur division showing lower disparities, though PM2.5 increased across all districts. Gazipur, Narsingdi, Narayanganj, and Dhaka consistently experienced the highest burden and the largest increases. Both air pollution and its associated health burden have risen and remain unevenly distributed, underscoring the need for targeted, region-specific interventions.

Keywords: Air pollution, PM2.5 Trends, LLP, Bangladesh.

Paper ID: 238

Socioeconomic Determinants of Cesarean Section Delivery in Bangladesh: A Survey-Weighted Statistical and Explainable Machine Learning AnalysisMd. Obaidul Islam¹, Md. Razu Ahmed¹, and Sabba Ruhi^{2*}¹Deep Statistical Learning and Research (DSLRL) Lab, Department of Statistics, Pabna University of Science and Technology, Pabna-6600²Department of Statistics, Pabna University of Science and Technology, Pabna-6600

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Abstract:

Cesarean section delivery in Bangladesh has reached 44.65% of all births, far above the World Health Organization's recommended population threshold of 10–15%, raising concerns about over-medicalization of childbirth and inequality in maternal healthcare. This study investigates the socioeconomic and healthcare-system determinants of cesarean section delivery using nationally representative data from the Bangladesh Demographic and Health Survey 2022. The final analytical sample comprised 5,052 women. The study applies a three-stage



analytical framework that combines survey-weighted generalized estimating equations for population-representative inference, six machine-learning classifiers with stratified cross-validation and Youden's J-based threshold optimization for prediction, and SHapley Additive exPlanations for global and individual-level interpretability. Bivariate analysis shows that delivery facility type is the strongest predictor, with cesarean prevalence reaching 84.3% in private facilities and remaining near zero in home deliveries (Cramér's $V = 0.763$). In the adjusted survey-weighted model, the richest wealth quintile is associated with increased odds of cesarean delivery (AOR = 1.78), four or more antenatal visits are associated with higher odds (AOR = 1.64), and fourth- or higher-order births are associated with lower odds (AOR = 0.48). Regional variation also remained significant after adjustment, indicating important differences in cesarean delivery. Across the predictive models, performance remains consistently strong, with the Stacking Ensemble achieving the best overall balance, including an F1-score of 87.06%, ROC-AUC of 0.924, and recall of 91.22%. SHAP analysis confirms delivery facility type, wealth status, and birth order as the most influential predictors and further shows that private facility delivery and intensive antenatal care contribute more strongly to cesarean risk among socioeconomically advantaged women. Overall, these findings indicate that cesarean section delivery in Bangladesh is shaped not only by obstetric factors but also by healthcare-system structure and socioeconomic inequality. The study highlights the need for stronger regulation of private maternity care, improved antenatal counselling, and equity-focused maternal health interventions.

Keywords: Cesarean Section; Survey-Weighted GEE; Explainable Machine Learning; Socioeconomic Inequality; Private Health Facilities.

Paper ID: 257

A Longitudinal NLP Framework for Predicting Scientific Manuscript Outcomes in the Revision Process

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Abstract

Peer review is the primary process through which scientific manuscripts are evaluated and revised prior to acceptance for publication. However, there exists a lack of clear understanding of which factors determine how much revision a manuscript requires. It is widely assumed that revision burden reflects identifiable weaknesses in the manuscript text itself, and that reviewers communicate high demand through explicit and direct criticism. This study tests these assumptions by utilizing a Natural Language Processing (NLP) based approach to investigate which signals in the reviewing stage drive revision demand at each cycle of the editorial process, and whether that demand can be correctly predicted from the manuscript before review begins. To do so, complete revision histories were reconstructed for 615 biomedical articles submitted to PeerJ, an open-access journal with public peer review records. This yielded a final dataset of 1,710 manuscript versions, each containing its metadata and full text. Each record further contained the reviewer comments it received for that version. Predictive features were then derived from this dataset, which were organized into three principal categories. These spanned manuscript-text features at submission (e.g., Flesch readability, statistical-term density), inter-version dynamics measuring edit magnitude (e.g., text growth, abstract-rewrite distance), and reviewer-discourse features from each round (e.g., uncertainty-phrase and major-critique counts, inter-reviewer semantic distance). After feature selection, the three categories comprised up to 89 features. An ablation study using XGBoost ordinal classifier with a five-fold cross-validation was trained across combinations of these feature sets to identify which signals best characterize revision demand at each review round. Additionally, L2-penalised logistic regression and Kaplan-Meier survival analysis were used to establish how accurately pre-review manuscript features can predict